

# Yuchen Wang

Agentic post-training | ML systems

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## Education

### University of Illinois Urbana-Champaign

Master of Computer Science

2025 – 2027 (expected)

Champaign, IL

### Peking University

B.S. in Intelligent Science and Technology (AI); Zhi Class (Elite Program), Top 10%

2021 – 2025

Beijing, China

## Internships

### Alibaba U.S. – Accio Team

Research Intern, LLM Training Algorithms & Infrastructure

May 2026 – Dec 2026

Sunnyvale, CA

- Core contributor to **Occamy-1.0** [arXiv] (35B-A3B); co-designed staged SFT/HDPO, model soup, and SAO with trainable context compaction and cross-segment credit assignment. **500K+** 📄 **downloads across Occamy variants**.
- Architected environment-/capability-first data synthesis and multi-harness training infrastructure with state-aware graders and asynchronous actor-critic learning; the deduplicated SFT union spans **15K trajectories and 403M tokens**.
- Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass **7.5% → 27.6%** and Claw-Eval average **69.5 → 82.2**, while reducing Claw-Eval T/C full-trial time **36.6%**.

### Freedo Technology

Research Intern, 3D Vision and Generative AI

May 2025 – Aug 2025

Beijing, China

- Raised point-cloud geometric fidelity from **32.7% to 85.3%** by conditioning ControlNet on depth and surface normals while preserving edges and planar surfaces.

## Open Source

- RL infrastructure:** bypassed the Ray driver for **NeMo RL** teacher top- $k$  transfers (**4.44–5.31×** in controlled  $8\times B200$  tests); corrected async GRPO/distillation checkpoint-resume bookkeeping in **TRL**.
- Long-context training:** developing **Qwen4-Exp** in Megatron Bridge; added **fused GDN Q/K normalization** and **Mamba recompute** in Megatron-LM; made **mc core-bridge** packed QSA selection **2.70×** **faster** on an 8K synthetic case.
- Inference:** halved 120B MoE TP2 BF16 conversion peak (**7.88 → 3.94 GiB**) in **vLLM**; fixed calibrated FP8 KV prefill in **FlashInfer** and Mamba cache-publication ordering in **SGLang**.
- Optimizer correctness:** repaired **Muon** small-gradient updates and Megatron-LM **checkpoint-safe Adam counters**.
- Training data:** preserved source sharding in **HF Datasets** streaming shuffle, restoring **all four workers on 19 shards**.

## Research

### RareDx: Graph-Grounded RL for Rare-Disease Diagnosis [arXiv]

Jun 2026 – Sep 2026

Co-first author; Advisor: Prof. Tianyu Liu, Tsinghua University

- Co-designed RL rewards for partial diagnostic matches using disease-ontology distance, biomedical similarity, and phenotype-disease links. With SFT and retrieval, 27B diagnosis Hit@10 increased from **28.53% to 40.76%**.

### CineFlow: Dependency-Driven Parallel Video Generation [Paper]

Oct 2025 – May 2026

Co-first author; Advisor: Prof. Fan Lai, University of Illinois Urbana-Champaign

- Developed a training-free, critical-path scheduler for video diffusion using semantic DAGs, trajectory fusion, and online edge pruning; analyzed 1K+ MovieGenBench prompts.
- Delivered **1.7–5.5×** **end-to-end speedup**, **1.30–2.02×** lower P90 latency, and **5.4–17.3%** **higher VBench overall** across Wan2.2-5B, CogVideoX-5B, and HunyuanVideo on  $8\times H100$ .

### Dynamic Prefill Optimization via Adaptive Online Packing [Report]

Aug 2025 – Jan 2026

University of Illinois Urbana-Champaign

- Built an AIMD batch-size controller driven by live p95 TTFT and packed mixed-length prompts with greedy/DP scheduling; cut TTFT by **up to 20%** on production-style DynamoLLM traces.

### FlashAttention-style CUDA Kernels for GPT-2 [Report]

Aug 2025 – Dec 2025

University of Illinois Urbana-Champaign

- Accelerated GPT-2 through tiled attention, online softmax, and kernel fusion, raising arithmetic intensity from **15 to 157 FLOP/byte**, cutting HBM traffic roughly **10×**, and improving end-to-end speed up to 9%.

### Reinforcement Learning for Legal Reasoning [Thesis]

Jan 2025 – Mar 2025

Advisor: Prof. Yansong Feng, Peking University

- Post-trained a legal model with Zero-RL, chain-of-thought-distilled SFT, and GRPO; reached **57.6% accuracy**, beating SFT-only and RL-only baselines.

## Service, Awards & Technical Skills

Reviewer, **WSDM 2027**

**Awards:** Merit Student; Zhi Class Scholarship; CMO (Provincial First Prize), 2020.

**Languages:** Python, C++, CUDA. **Systems:** PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight.

**Methods:** RL, distributed training, diffusion transformers, ControlNet, 3D vision and point clouds.